PRODUCT AND TECHNICAL ARCHITECTURE

Enterprise Real-Time Voice-to-Workflow Platform

A configurable enterprise platform that converts live Arabic speech into validated, secure, executable business workflows.

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This document defines the product vision, system boundaries, real-time architecture, security model, API contracts, delivery roadmap, and acceptance criteria.

Contents

1 Executive Summary

Executive Summary

Enterprise Real-Time Voice-to-Workflow Platform is a multi-tenant enterprise platform that allows each organization to define its own business workflows, data schemas, terminology, validation rules, permissions, and integration targets.
An employee can speak naturally in Arabic or Egyptian Arabic while the platform processes the speech as a real-time stream. The system produces a continuously updated structured workflow draft, validates it against the organization's configuration, asks for missing information when necessary, and executes the final action only after the configured confirmation and approval requirements are satisfied.
The platform is not only a speech-to-text service and is not a generic large-language-model wrapper. Its primary value is the combination of:

- Real-time streaming speech processing.
- Tenant-defined JSON schemas and workflow definitions.
- Incremental structured-data extraction.
- Deterministic business validation.
- Enterprise identity and permission enforcement.
- Secure workflow execution.
- Webhooks, APIs, queues, and n8n integrations.
- Auditing, replay, observability, and quality evaluation.

2 Problem Definition

Operational employees frequently communicate business events using voice because typing structured forms is slow, inconvenient, or impractical in field environments.
Examples include:

- Drivers recording sales, purchases, collections, or expenses.
- Warehouse employees recording stock movements.
- Sales representatives recording customer orders.
- Maintenance teams recording incidents and completed work.
- Recruitment teams updating candidate or worker states.
- Healthcare and service teams recording operational notes.

A typical spoken statement may contain multiple entities, calculations, corrections, and business decisions. For example:

> اشتريت من شركة النور ثلاثة طن أرز، الطن بأربعة وعشرين ألف، ودفعت ثلاثين ألف كاش، والباقي آجل، والنقل ألف ومائتين جنيه.

Traditional speech-to-text systems return plain text. They do not reliably determine:

- Which business workflow the employee intends to execute.
- Which fields belong to the workflow.

- Whether the values comply with company rules.
- Whether referenced suppliers, customers, products, or accounts exist.
- Whether the employee is authorized to perform the action.
- Whether confirmation or approval is required.
- Which external application should receive the final result.

3 Product Vision

3.1 Vision Statement

Product Vision

Provide enterprises with a configurable runtime that converts natural spoken language into validated business actions without requiring each organization to build its own speech, AI, workflow, security, and integration infrastructure.

3.2 Product Positioning

The platform should be positioned as:

Enterprise Real-Time Voice Workflow Infrastructure

It should not be positioned as:

- A generic transcription tool.
- A chatbot builder.
- A fixed accounting application.
- A general-purpose automation platform.
- An unvalidated AI action executor.

3.3 Core Differentiation

The platform's competitive advantage is the combination of:

1. Streaming voice processing with low perceived latency.
2. Configurable enterprise workflows.
3. Constrained JSON output.
4. Deterministic validation and entity resolution.
5. Secure and auditable execution.
6. Arabic and Egyptian Arabic operational language support.

4 Target Customers and Use Cases

4.1 Primary Customer Segments

Segment	Operational Need	Example Workflows
Distribution companies	Fast field data capture	Sales, collections, expenses, stock transfer
Agricultural companies	Field and warehouse operations	Purchase, delivery, batch movement, inspection
Logistics companies	Driver and shipment reporting	Delivery, incident, fuel expense, cash collection
ERP and accounting vendors	Voice-enabled transaction entry	Purchase, sale, payment, receipt, journal draft
Maintenance companies	Hands-free field reporting	Work order, incident, parts used, completion report
Recruitment agencies	Worker and candidate workflow updates	Candidate status, document collection, assignment
Healthcare operations	Structured operational recording	Visit notes, inventory usage, service completion

4.2 Primary Personas

- **Enterprise Administrator:** defines workflows, schemas, rules, integrations, and access policies.
- **Operational Employee:** speaks naturally and reviews the generated structured action.
- **Supervisor or Approver:** reviews high-risk actions before execution.
- **Integration Engineer:** connects the platform to internal systems, APIs, and n8n.
- **Security Administrator:** manages identity, keys, retention, access controls, and audit policies.
- **Quality Analyst:** reviews extraction quality, failed sessions, corrections, and model metrics.

5 Core Product Concepts

5.1 Tenant

A tenant represents one customer organization. All workflow definitions, data, credentials, logs, terminology, and integrations are isolated by tenant.

5.2 Workflow Definition

A workflow definition describes a business operation such as:

- Purchase
- Sale
- Income
- Expense
- Receipt
- Payment
- Stock transfer

- Maintenance request
- Candidate status update

Each workflow contains:

- Unique key and display name.
- Business description.
- JSON Schema.
- Required and optional fields.
- Trigger phrases and terminology.
- Examples.
- Validation rules.
- Entity-resolution rules.
- Permission requirements.
- Execution policy.
- Integration destination.
- Version.

5.3 Voice Session

A voice session represents one continuous employee interaction. It stores:

- Tenant and user identity.
- Audio stream metadata.
- Transcript revisions.
- Workflow candidates.
- Current structured draft.
- Missing fields.
- Validation results.
- Confirmation state.
- Final execution result.

5.4 Workflow Draft

A workflow draft is an incomplete or complete structured representation of the employee's spoken intent. It is mutable while the employee is speaking and immutable after final commit.

5.5 Execution Policy

The execution policy controls what must happen before the platform sends or executes the final workflow. Supported modes should include:

- Automatic execution.
- User confirmation.

- PIN or biometric confirmation.
- Supervisor approval.
- Dual approval.
- Draft creation only.

6 End-to-End User Experience

6.1 Administrator Experience

The enterprise administrator opens the dashboard and performs the following process:

1. Creates a new workflow.
2. Defines its name, description, and business purpose.
3. Creates or imports its JSON Schema.
4. Marks required and optional fields.
5. Adds example phrases.
6. Defines company-specific synonyms.
7. Adds validation rules.
8. Configures database or API entity resolution.
9. Assigns roles and permissions.
10. Selects an execution policy.
11. Connects a webhook, API, queue, or n8n workflow.
12. Tests the workflow in a voice playground.
13. Publishes a version for production use.

6.2 Employee Experience

The operational employee:

1. Opens the voice-enabled application.
2. Starts a secure voice session.
3. Speaks naturally.
4. Sees an immediate partial transcript.
5. Sees a structured draft update while speaking.
6. Corrects information naturally when necessary.
7. Provides missing information when asked.
8. Reviews the final structured result.
9. Confirms the action.
10. Receives a success, rejection, or approval-pending result.

6.3 Example Interaction

The employee says:

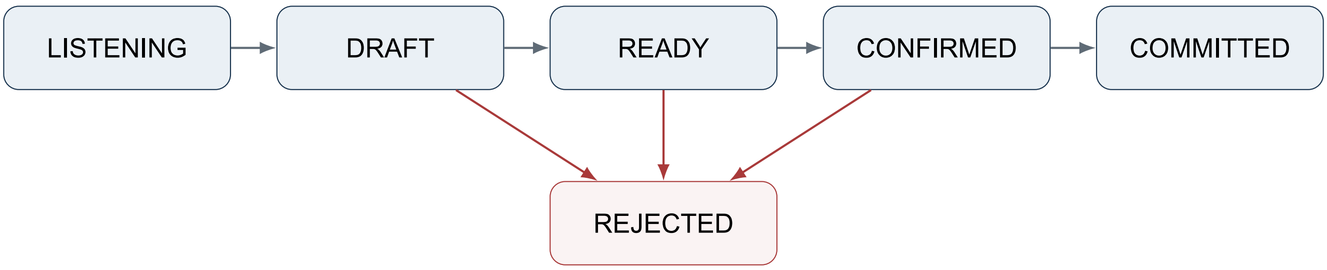
بعت للعميل أحمد ثلاثة طن أرز، الطن باثنين وعشرين ألف، واستلمت منه عشرة آلاف كاش، والباقي آجل.

The platform incrementally identifies:

- Workflow: Sale
- Customer: Ahmed
- Product: Rice
- Quantity: 3
- Unit: Ton
- Unit price: 22,000
- Paid amount: 10,000
- Remaining amount: 56,000
- Payment status: Partial

7 Workflow Lifecycle

The workflow session follows a controlled lifecycle.



State	Meaning
LISTENING	Audio is being received and transcribed.
DRAFT	The platform has identified a possible workflow and partial fields.
READY	Required information is complete and deterministic validation has passed.
CONFIRMED	The user or approver has confirmed execution.
COMMITTED	The final action has been delivered or executed successfully.
REJECTED	The action was rejected due to validation, authorization, approval, or integration failure.

Critical Execution Rule

Partial transcripts and draft updates must never directly trigger irreversible business actions.
Only a confirmed and validated final workflow may reach the execution dispatcher.

8 Real-Time Processing Architecture

8.1 High-Level Architecture

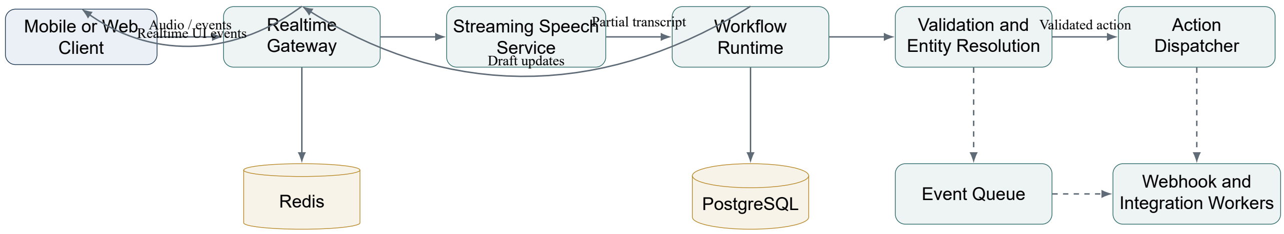


Figure 1: High-level real-time and asynchronous processing architecture.

8.2 Realtime Gateway

The Realtime Gateway is responsible for:

- WebSocket or WebRTC session management.
- Authentication and tenant resolution.
- Audio chunk sequencing.
- Backpressure and connection health.
- Reconnection and session recovery.
- Rate limiting and abuse prevention.
- Routing sessions to speech-processing workers.
- Returning transcript and workflow events to the client.

The gateway should remain stateless where possible. Hot session state should be stored in Redis.

8.3 Streaming Speech Service

The speech service is responsible for:

- Audio format validation.
- Voice activity detection.
- Noise and silence handling.
- Streaming speech-to-text.
- Partial transcript generation.
- Final transcript generation.
- Transcript stability scoring.
- Arabic number normalization.
- Egyptian Arabic terminology normalization.

8.4 Workflow Runtime

The Workflow Runtime is responsible for:

- Loading tenant workflow definitions.

- Selecting workflow candidates.
- Extracting structured fields.
- Updating the current draft.
- Handling corrections and revisions.
- Tracking source spans and confidence.
- Identifying missing required fields.
- Deciding when deeper model reasoning is required.

8.5 Validation and Entity Resolution

This service provides deterministic validation after AI extraction.
It validates:

- JSON Schema compliance.
- Required fields.
- Numeric ranges.
- Cross-field business rules.
- Entity existence.
- User permissions.
- Tenant restrictions.
- Duplicate-action risks.
- Accounting and inventory constraints.

8.6 Action Dispatcher

The Action Dispatcher executes only finalized workflows.
Supported delivery mechanisms should include:

- Signed webhook delivery.
- REST API invocation.
- Message queue publication.
- n8n webhook trigger.
- Internal connector invocation.
- Draft-only persistence.

9 Low-Latency Processing Strategy

9.1 Fast Layer and Reasoning Layer

The system should not send every partial word to a large language model.
Instead, it should use two layers.

9.1.1 Fast Layer

The fast layer runs continuously and performs:

- Partial transcription.
- Number normalization.
- Keyword recognition.
- Basic entity extraction.
- Candidate workflow detection.
- Session state updates.

9.1.2 Reasoning Layer

The reasoning layer runs at semantic checkpoints, including:

- A stable phrase or sentence boundary.
- A short period of silence.
- Detection of a new business entity.
- Detection of a correction.
- Completion of required workflow information.
- Ambiguity requiring structured reasoning.
- Final end-of-speech processing.

9.2 Target Latency Budget

Stage	**Target**	**User-Visible Result**
Audio chunk upload	20–100 ms	Continuous connection
Partial speech recognition	100–300 ms	Live transcript
Fast entity extraction	10–60 ms	Immediate field updates
Draft UI update	Below 100 ms	Responsive structured form
Structured model reasoning	200–900 ms	Corrected workflow draft
Final deterministic validation	20–250 ms	Ready or validation-error state

These figures are architectural targets and should be verified through load testing against the selected speech and language models.

10 Workflow Configuration Model

10.1 Example Workflow Definition

Listing 1: Example tenant workflow definition

```
{
"key": "purchase",
"version": 3,
"name": "Purchase",
"description": "Record the purchase of goods from a supplier",
"status": "published",

"triggerPhrases": [
```

```
"I purchased",
"we bought",
"goods received from supplier"
],

"schema": {
"type": "object",
"required": ["supplierId", "items"],
"properties": {
"supplierId": {
"type": "string"
},
"items": {
"type": "array",
"minItems": 1,
"items": {
"type": "object",
"required": [
"productId",
"quantity",
"unitId",
"unitPrice"
],
"properties": {
"productId": {
"type": "string"
},
"quantity": {
"type": "number",
"exclusiveMinimum": 0
},
"unitId": {
"type": "string"
},
"unitPrice": {
"type": "number",
"minimum": 0
}
}
}
},
"paidAmount": {
"type": "number",
"minimum": 0
},
"paymentMethod": {
"type": "string",
"enum": ["cash", "bank", "credit"]
},
"expenses": {
"type": "array"
}
}
},

"rules": [
{
"key": "payment_not_above_total",
"expression": "paidAmount <= invoiceTotal",
"severity": "error"
}
],

"executionPolicy": {
"mode": "confirm",
"requiresPin": false,
```

```
"requiresSupervisorApproval": false
},

"integration": {
"type": "webhook",
"destinationId": "dest_01KEXAMPLE"
}
}
```


10.2 Workflow Versioning

Workflow definitions must be versioned.
Publishing a new version must not alter historical sessions created using older versions.
Each session must store:

- Workflow definition ID.
- Workflow version.
- Model configuration version.
- Validation-rules version.
- Integration-destination version.

10.3 Schema Builder

The dashboard should provide both:

- A visual schema builder for non-technical administrators.
- An advanced JSON Schema editor for engineers.

The visual builder should support:

- Text, number, boolean, date, enum, object, and array fields.
- Required and optional fields.
- Default values.
- Numeric limits.
- String patterns.
- Field descriptions.
- Entity-reference fields.
- Conditional fields.
- Repeatable line items.

11 Realtime Protocol

11.1 Connection Establishment

The client creates a session using a standard authenticated API request, then opens a realtime connection.

Listing 2: Create a voice session

```
POST /v1/realtime/sessions

```

```
{
"locale": "ar-EG",
"allowedWorkflowKeys": [
"purchase",
"sale",
"expense",
"income"
],
"audio": {
"format": "pcm16",
"sampleRate": 16000,
"channels": 1
}
}
```


11.2 Client Audio Event

Binary frames are preferred for audio payloads to avoid Base64 overhead.

A corresponding metadata event may use the following structure:

Listing 3: Audio chunk metadata event


```
{
"type": "audio.chunk",
"sequence": 142,
"durationMs": 100,
"timestamp": "2026-07-23T18:30:11.420Z"
}
```


11.3 Partial Transcript Event

Listing 4: Partial transcript event


```
{
"type": "transcript.partial",
"sessionId": "vs_01KEXAMPLE",
"revision": 12,
"text": "I purchased three tons of rice",
"isStable": false,
"confidence": 0.82
}
```


11.4 Workflow Draft Update

Listing 5: Incremental workflow draft update


```
{
"type": "workflow.draft.updated",
"sessionId": "vs_01KEXAMPLE",
"revision": 7,
"workflow": {
"key": "purchase",
"state": "selected",
"confidence": 0.91
},
"data": {
"supplierId": "supplier_102",
"items": [
{
"productId": "product_rice_01",
"quantity": 3,
```

```
"unitId": "ton"
}
]
},
"missingFields": [
{
"path": "items[0].unitPrice",
"reason": "required"
}
]
}
```

11.5 Final Workflow Event

Listing 6: Workflow ready for confirmation

```
{
"type": "workflow.ready",
"sessionId": "vs_01KEXAMPLE",
"revision": 14,
"workflowKey": "purchase",
"data": {
"supplierId": "supplier_102",
"items": [
{
"productId": "product_rice_01",
"quantity": 3,
"unitId": "ton",
"unitPrice": 24000,
"lineTotal": 72000
}
],
"paidAmount": 30000,
"remainingAmount": 42000,
"paymentMethod": "cash"
},
"validation": {
"valid": true,
"errors": [],
"warnings": []
},
"confirmationRequired": true
}
```

12 Corrections and Transcript Revisions

Employees frequently correct themselves while speaking.

Example:

بعت خمسة طن... لا، ثلاثة طن.

The system must support revision-based updates rather than append-only transcription.

Listing 7: Transcript correction event

```
{
"type": "transcript.revision",
"revision": 18,
"replacement": {
"startToken": 4,
"endToken": 6,
```

```
"oldText": "five tons",
"newText": "three tons"
}
}
```

The Workflow Runtime should then update the affected field.

Listing 8: Structured field correction

```
{
"type": "workflow.field.corrected",
"revision": 9,
"path": "items[0].quantity",
"oldValue": 5,
"newValue": 3,
"reason": "user_correction",
"confidence": 0.96
}
```

Each extracted field should retain provenance metadata:

- Transcript source range.
- Transcript revision.
- Extraction method.
- Confidence.
- Last updated timestamp.
- Whether the value was user-confirmed.

13 Entity Resolution

The platform should not assume that extracted names represent valid business entities.

For example, “Ahmed” may match multiple customers.

Entity resolution should support:

- Exact matching.
- Alias matching.
- Phonetic matching.
- Fuzzy text matching.
- Recent-entity prioritization.
- User-assigned entity scope.
- External API lookup.
- Human clarification.

When multiple entities are possible, the system should return candidates instead of silently selecting one.

Listing 9: Entity ambiguity event

```
{
"type": "workflow.clarification.required",
"field": "customerId",
"question": "Which Ahmed customer do you mean?",
"candidates": [
{
"id": "customer_11",
```

```
"label": "Ahmed Hassan - Cairo",
"confidence": 0.78
},
{
"id": "customer_42",
"label": "Ahmed Ali - Tanta",
"confidence": 0.72
}
]
}
```


14 Validation Model

14.1 Validation Layers

Validation should occur through multiple layers.

1. **Schema Validation**

   Confirms that the output structure and field types match the published JSON Schema.

2. **Business Rule Validation**

   Evaluates rules such as payment limits, quantity limits, account state, and workflow-specific constraints.

3. **Entity Validation**

   Confirms that referenced customers, suppliers, products, warehouses, and accounts exist and are active.

4. **Authorization Validation**

   Confirms that the authenticated employee may create, confirm, approve, or execute the workflow.

5. **Risk Validation**

   Detects duplicate actions, unusual values, policy violations, or actions requiring additional approval.

6. **Integration Validation**

   Confirms that the destination configuration is active and compatible with the final payload version.

14.2 Validation Error Structure

Listing 10: Structured validation error


```
{
"valid": false,
"errors": [
{
"code": "PAID_AMOUNT_EXCEEDS_TOTAL",
"path": "paidAmount",
"message": "Paid amount cannot exceed the invoice total",
"severity": "error",
"source": "business_rule"
}
],
"warnings": []
}
```

15 Security Architecture

Security Principle

The language model must never be treated as an authorization system, validation authority, or trusted execution engine.

15.1 Identity and Access Management

The platform should support:

- OpenID Connect and SAML for enterprise identity.
- JWT access tokens with short expiration.
- Refresh-token rotation.
- Multi-factor authentication.
- Role-based access control.
- Attribute-based policies for sensitive operations.
- Device registration and revocation.
- Service accounts for integrations.

15.2 Tenant Isolation

Tenant isolation must be enforced at:

- API authorization.
- Database queries.
- Redis keys.
- Queue messages.
- Object-storage paths.
- Encryption context.
- Metrics and logs.
- Model prompts and context.

Every internal request should include a verified tenant context. Tenant identifiers received from clients must not be trusted without authentication and authorization.

15.3 Data Protection

The system should implement:

- TLS for all external and internal traffic.
- Encryption at rest.
- Tenant-specific encryption keys for enterprise plans.
- Secret management outside application code.
- Configurable audio-retention policies.

- Transcript redaction.
- Personally identifiable information controls.
- Secure deletion workflows.

15.4 Webhook Security

Webhook delivery should support:

- HMAC signatures.
- Timestamped requests.
- Replay-attack protection.
- Secret rotation.
- IP allowlists.
- Mutual TLS for enterprise customers.
- Idempotency keys.
- Retry and dead-letter policies.
- Delivery logs.

15.5 Prompt and Model Security

The system must protect against:

- Prompt injection in spoken content.
- Attempts to override workflow constraints.
- Unauthorized workflow selection.
- Data exfiltration through model responses.
- Cross-tenant context leakage.
- Malicious entity names or external content.

  The model should receive only:

- The workflows allowed for the current user.
- The minimum required tenant terminology.
- The current transcript window.
- The current structured draft.
- Explicit system-controlled output constraints.

16 Auditability and Compliance

Every session should produce a complete audit record containing:

- Tenant ID.
- User ID.
- Device ID.

- Session timestamps.
- Workflow definition and version.
- Transcript revisions.
- Draft revisions.
- Model and prompt versions.
- Validation results.
- Confirmation evidence.
- Approval decisions.
- Final payload.
- Integration-delivery attempts.
- Final execution outcome.

Audio storage must be configurable per tenant:

- Do not store.
- Store temporarily.
- Store only failed sessions.
- Store with explicit employee consent.
- Store according to customer-defined retention periods.

17 Scalability and Availability

17.1 Independent Scaling Units

The following components should scale independently:

- Realtime Gateway: concurrent connections.
- Speech Workers: concurrent audio streams.
- Workflow Runtime: extraction and reasoning requests.
- Validation Workers: database and external lookups.
- Webhook Workers: delivery throughput.
- Audit Pipeline: event-ingestion volume.

17.2 Hot and Durable State

Storage	Purpose	Examples
Redis	Short-lived realtime state	Session state, connection routing, rate limits
PostgreSQL	Durable relational state	Tenants, workflows, sessions, approvals, audit metadata
Object storage	Large binary objects	Audio, attachments, exported audit packages
Event queue	Durable asynchronous processing	Webhook jobs, analytics, quality evaluation

17.3 Failure Handling

The architecture should support:

- Realtime reconnection using session resume tokens.
- Audio sequence-number verification.
- Duplicate chunk detection.
- Idempotent finalization.
- At-least-once asynchronous delivery.
- Idempotency keys for downstream execution.
- Dead-letter queues.
- Controlled retry with exponential backoff.
- Circuit breakers for external dependencies.

18 Suggested Data Model

Entity	Purpose	Important Fields
Tenant	Customer organization	ID, name, plan, region, security policy
User	Human platform user	Tenant, identity provider, roles, status
Device	Registered client device	User, fingerprint, status, last seen
WorkflowDefinition	Logical workflow	Tenant, key, name, current published version
WorkflowVersion	Immutable workflow version	Schema, rules, examples, execution policy
TerminologyEntry	Tenant vocabulary	Phrase, normalized value, entity type
IntegrationDestination	External action destination	Type, endpoint, secret reference, status
VoiceSession	Realtime interaction	User, workflow candidate, status, timestamps
TranscriptRevision	Transcript history	Session, revision, text, stability, confidence
WorkflowDraftRevision	Structured draft history	Session, revision, payload, missing fields
ValidationResult	Validation outcome	Session, rule, severity, field path
Confirmation	User confirmation evidence	User, method, timestamp, device
Approval	Supervisor decision	Approver, status, reason, timestamp
ExecutionAttempt	Delivery or execution attempt	Destination, attempt number, response, status
AuditEvent	Immutable operational audit	Actor, action, target, timestamp, metadata

19 API Surface

19.1 Management APIs

- `POST /v1/workflows`
- `GET /v1/workflows`
- `GET /v1/workflows/workflowId`
- `POST /v1/workflows/workflowId/versions`

- POST /v1/workflows/workflowId/publish
- POST /v1/workflows/workflowId/test
- POST /v1/integrations
- POST /v1/terminology
- GET /v1/quality/sessions

19.2 Realtime APIs

- POST /v1/realtime/sessions
- GET /v1/realtime/sessions/sessionId
- POST /v1/realtime/sessions/sessionId/finalize
- POST /v1/realtime/sessions/sessionId/confirm
- POST /v1/realtime/sessions/sessionId/cancel
- WSS /v1/realtime/sessions/sessionId/stream

19.3 Approval APIs

- GET /v1/approvals
- POST /v1/approvals/approvalId/approve
- POST /v1/approvals/approvalId/reject

19.4 Webhook Event Types

- voice.session.started
- voice.session.completed
- workflow.draft.ready
- workflow.confirmed
- workflow.approval.requested
- workflow.approved
- workflow.rejected
- workflow.execution.succeeded
- workflow.execution.failed

20 Enterprise Dashboard

The dashboard should include the following modules.

20.1 Overview

- Active sessions.
- Completed workflows.
- Failed workflows.

- Average end-to-end latency.
- Extraction accuracy.
- Confirmation rate.
- Approval backlog.
- Integration-delivery health.

20.2 Workflow Builder

- Workflow details.
- Visual JSON Schema builder.
- Advanced JSON editor.
- Trigger phrases.
- Example utterances.
- Business terminology.
- Validation rules.
- Execution policy.
- Integration target.
- Version history.

20.3 Voice Playground

The Voice Playground should allow an administrator to:

- Record or upload voice.
- View partial transcripts.
- Inspect workflow candidates.
- Inspect extracted fields and confidence.
- Test corrections.
- Review missing-field questions.
- View validation results.
- Compare expected and actual JSON.
- Save examples to the evaluation dataset.

20.4 Quality Center

The Quality Center should provide:

- Failed-session review.
- Low-confidence sessions.
- Frequently corrected fields.
- Workflow confusion matrix.
- Terminology gaps.

- Model-version comparison.
- Regression-evaluation reports.

20.5 Integrations

- Webhook destinations.
- API connectors.
- n8n templates.
- Secret rotation.
- Delivery logs.
- Replay controls.
- Test-event delivery.

21 Observability

21.1 Technical Metrics

The platform should monitor:

- Active realtime connections.
- Audio-stream duration.
- Speech-service latency.
- Partial transcript latency.
- Workflow extraction latency.
- Validation latency.
- Model token usage.
- Queue depth.
- Webhook success rate.
- Database and Redis performance.

21.2 Product Quality Metrics

- Workflow classification accuracy.
- Field extraction precision.
- Field extraction recall.
- Correction frequency.
- Missing-field frequency.
- User confirmation rate.
- User cancellation rate.
- Manual-edit rate.
- Successful execution rate.

21.3 Tracing

Each session should have a distributed trace ID propagated through:

- Client.
- Realtime Gateway.
- Speech Service.
- Workflow Runtime.
- Validation Service.
- Dispatcher.
- Webhook Worker.

22 Deployment Topology

A production deployment may use:

- Kubernetes or another container orchestration platform.
- Regional realtime gateways.
- Dedicated speech-worker pools.
- Autoscaled workflow-runtime workers.
- Managed PostgreSQL with high availability.
- Redis with replication or clustering.
- Durable event streaming or queues.
- S3-compatible object storage.
- Centralized secrets management.
- OpenTelemetry-compatible observability.

  Enterprise deployment options may include:

- Multi-tenant managed cloud.
- Dedicated tenant environment.
- Private cloud.
- Customer-controlled self-hosted deployment.
- Hybrid edge and cloud deployment.

23 MVP Scope

23.1 Included in the MVP

The first commercially testable version should include:

- Multi-tenant organization management.
- User authentication and role-based access.
- Workflow creation.

- JSON Schema configuration.
- Workflow examples and terminology.
- Four initial workflow types: purchase, sale, income, and expense.
- Arabic and Egyptian Arabic streaming transcription.
- WebSocket realtime sessions.
- Incremental workflow drafts.
- Missing-field detection.
- Deterministic validation.
- User confirmation.
- Signed webhook delivery.
- n8n-compatible webhook templates.
- Session and delivery audit logs.
- Voice testing playground.
- Basic quality and latency dashboard.

23.2 Excluded from the MVP

The following should be postponed unless required by a paying pilot customer:

- Full offline speech recognition.
- Customer-specific model fine-tuning.
- On-premise installation.
- Advanced visual rule engine.
- Multi-step workflow orchestration.
- Marketplace of public workflow templates.
- Automated financial posting.
- Autonomous execution of high-risk actions.

24 Delivery Roadmap

Phase	Objectives	Main Deliverables
Phase 0: Validation	Confirm customer problem and technical feasibility.	Pilot workflows, voice samples, latency benchmark, security assumptions.
Phase 1: Foundation	Build tenant, IAM, workflow configuration, and schema versioning.	Management APIs, dashboard foundation, PostgreSQL model.
Phase 2: Realtime Runtime	Build secure streaming voice sessions.	Realtime Gateway, speech integration, Redis session state.
Phase 3: Structured Extraction	Convert partial transcripts into workflow drafts.	Workflow classification, field extraction, correction handling.
Phase 4: Validation	Add deterministic business validation and entity resolution.	Validation engine, lookup connectors, missing-field handling.
Phase 5: Execution	Safely deliver confirmed workflows.	Webhook dispatcher, retries, signatures, n8n templates.
Phase 6: Enterprise Readiness	Improve auditability, quality, scale, and security.	Quality center, tracing, retention policies, load tests.

25 Team Responsibilities

Team or Role	Responsibilities
Product Management	Customer discovery, workflow priorities, pricing, acceptance criteria.
Backend Engineering	APIs, tenancy, workflow runtime, validation, integrations, persistence.
Realtime Engineering	WebSocket/WebRTC gateway, audio streaming, session recovery, latency.
AI and ML Engineering	Speech models, extraction strategy, evaluation, model routing.
Frontend Engineering	Enterprise dashboard, schema builder, voice playground, session UI.
Security Engineering	Threat modeling, IAM, tenant isolation, keys, audit, compliance.
DevOps and SRE	Deployment, autoscaling, observability, reliability, incident response.
QA Engineering	Functional tests, realtime tests, regression datasets, load tests.
Domain Experts	Business rules, terminology, workflow examples, validation review.

26 Testing Strategy

26.1 Functional Testing

- Workflow configuration.
- Schema validation.
- Workflow version publishing.
- Permission enforcement.
- Session creation and completion.
- Confirmation and approval.
- Webhook delivery and replay.

26.2 Voice and AI Evaluation

The evaluation dataset should include:

- Egyptian Arabic accents.
- Background noise.
- Slow and fast speakers.
- Corrections.
- Multiple line items.
- Similar customer and product names.
- Mixed Arabic and English terminology.
- Spoken numbers and monetary amounts.
- Incomplete statements.
- Ambiguous workflows.

26.3 Performance Testing

Performance testing should cover:

- Concurrent WebSocket sessions.
- Long-running audio streams.
- Reconnection storms.
- Speech-worker saturation.
- Model-provider throttling.
- Redis failover.
- Queue backlog.
- Webhook destination failures.

26.4 Security Testing

- Cross-tenant access attempts.
- Token theft and replay.
- Prompt-injection attempts.
- Webhook forgery.
- Secret leakage.
- Malicious audio and payload sizes.
- Permission bypass attempts.
- Sensitive-data logging.

27 MVP Acceptance Criteria

The MVP may be considered ready for a controlled customer pilot when:

1. An enterprise administrator can define and publish a workflow without modifying backend code.
2. The administrator can provide a JSON Schema, examples, terminology, and validation rules.
3. An authenticated employee can start a realtime voice session.
4. Partial transcript updates are displayed while the employee speaks.
5. The system incrementally identifies one of the configured workflows.
6. The system produces a structured draft matching the published schema.
7. Spoken corrections update the existing draft instead of creating duplicate actions.
8. Missing required fields are identified and requested.
9. Deterministic validation prevents invalid final actions.
10. High-risk workflows cannot execute without their configured confirmation or approval.
11. Confirmed workflows are delivered through signed, idempotent webhooks.
12. Failed webhook deliveries are retried and visible in the dashboard.
13. Every final action is traceable through an immutable audit record.
14. Tenant data cannot be accessed by another tenant.
15. The realtime system satisfies the agreed pilot latency and concurrency targets.

28 Primary Risks and Mitigations

Risk	Impact	Mitigation
Poor dialect recognition	Incorrect transcript and extraction	Tenant terminology, evaluation datasets, model routing, confirmations
High model latency	Unresponsive experience	Fast extraction layer, semantic checkpoints, caching, regional workers
Incorrect action execution	Financial or operational damage	Draft-only processing, deterministic validation, confirmation, approval
Cross-tenant leakage	Critical security breach	Strict tenant context, row-level enforcement, isolated keys and logs
Prompt injection	Invalid or unauthorized model behavior	Constrained outputs, limited context, independent authorization
External integration failure	Lost or duplicate actions	Idempotency, retries, dead-letter queues, delivery logs
Cost growth	Unsustainable margins	Model routing, audio limits, batching, per-tenant quotas, usage pricing
Workflow complexity	Difficult enterprise onboarding	Visual builder, templates, guided testing, professional services

Risk	Impact	Mitigation
Noisy environments	Poor speech quality	Voice activity detection, noise handling, client microphone guidance
User distrust	Low adoption	Live draft preview, explainable corrections, final confirmation

29 Commercial Model

A hybrid commercial model is recommended.

29.1 Platform Subscription

The subscription may be based on:

- Number of active users.
- Number of voice minutes.
- Number of completed workflows.
- Number of published workflow definitions.
- Data-retention requirements.
- Deployment model.

29.2 Usage Charges

Usage charges may cover:

- Streaming speech minutes.
- Structured-model processing.
- Webhook events.
- Stored audio.
- Premium entity-resolution connectors.

29.3 Enterprise Services

Additional revenue may come from:

- Workflow design.
- Integration implementation.
- Custom deployment.
- Security review.
- Domain-specific terminology preparation.
- Evaluation-dataset preparation.
- Service-level agreements.

30 Recommended Initial Pilot

The first pilot should focus on one operational customer and four high-frequency workflows:

1. Purchase
2. Sale
3. Income or receipt
4. Expense or payment

The pilot should avoid direct irreversible accounting posting.

The initial integration should create a validated transaction draft or send a signed webhook to the customer's backend. The customer's system remains the final system of record.

This approach reduces risk while proving:

- Real-time usability.
- Arabic speech quality.
- Workflow configurability.
- Integration value.
- Operational time savings.
- Customer willingness to pay.

31 Final Recommendation

The project should be developed as a secure, multi-tenant, configuration-driven runtime rather than as a collection of hard-coded voice commands.

The architecture should preserve a strict separation between:

- Speech recognition.
- AI-based extraction.
- Deterministic validation.
- Authorization.
- Confirmation and approval.
- Final execution.

The first product milestone should prove one complete vertical flow:

Tenant Workflow Definition → Realtime Voice Session → Structured Draft → Validation → User Confirmation → Signed Webhook

Once this flow is stable, measurable, and secure, the platform can expand into additional industries, languages, deployment models, and workflow types.

A Decision Summary

Decision	Selected Direction
Primary product category	Enterprise realtime voice workflow infrastructure
Customization model	Tenant-defined schemas, examples, terminology, and rules
Realtime transport	WebSocket initially; WebRTC where audio requirements justify it
Execution model	Streaming draft, final confirmed execution
State model	Redis for hot state, PostgreSQL for durable state
Integration model	Signed webhooks, REST APIs, queues, and n8n
AI role	Classification and extraction, not authorization or final validation
MVP workflows	Purchase, sale, income, and expense
Initial delivery risk	Create validated drafts or webhooks, not direct financial posting
Security model	Tenant isolation, RBAC, audit, encryption, signed delivery

B Glossary

Tenant A customer organization isolated within the platform.

Workflow Definition A versioned configuration describing a business operation and its structured data.

Workflow Draft The mutable structured result produced while the employee is speaking.

Voice Session A realtime interaction containing audio, transcripts, drafts, validations, and execution state.

Entity Resolution Mapping spoken names to valid customer, supplier, product, account, or other business identifiers.

Execution Policy Rules controlling confirmation, authentication, approval, and final action execution.

Semantic Checkpoint A meaningful point at which the reasoning model updates the structured workflow draft.

System of Record The authoritative customer application that owns the final business transaction.